



SYSTEM OVERVIEW

AUTOVET · VERSION 1.0

SYSTEM DESCRIPTION

AutoVet is a hybrid veterinary clinic management system that unites a cloud web portal, an offline-capable clinic workstation, and AI-assisted forecasting into a single platform. Pet owners book appointments and manage pet profiles online, while clinic staff handle patients, walk-ins, scheduling, billing, and stock from one dashboard.

Built on a Laravel REST API with React front-ends, AutoVet keeps the clinic running even when connectivity drops and resynchronizes securely once it returns — replacing paper logbooks and spreadsheets with role-based, fully auditable digital records.

OBJECTIVES

- Automate appointment booking, scheduling, and clinic transactions.
- Centralize pet profiles, medical histories, and vaccination records.
- Prevent stockouts through FIFO batch tracking and AI demand forecasting.
- Generate invoices, receipts, and management reports instantly.
- Secure patient and owner data with role-based access and audit logging.
- Sustain clinic operations offline and sync reliably to the cloud.

3

SYSTEM LAYERS

5

USER ROLES

16

CORE MODULES

187

API ENDPOINTS



SYSTEM FEATURES

Clinical Operations

PART 1 OF 3

APPOINTMENTS & SCHEDULING

- ✓ Online slot-based booking
- ✓ Walk-in registration
- ✓ Veterinarian schedules
- ✓ Status workflow tracking
- ✓ Email confirmations
- ✓ Double-booking prevention
- ✓ Calendar & day views
- ✓ Availability management
- ✓ Approval & cancellation
- ✓ Automated reminders

PATIENTS & MEDICAL RECORDS

- ✓ Pet profiles with photos
- ✓ Electronic medical records
- ✓ Vaccination tracking
- ✓ Auto age-group from DOB
- ✓ Species & breed catalogue
- ✓ Owner & client records
- ✓ Treatment & visit history
- ✓ Allergies & medication notes
- ✓ Auto size from weight range
- ✓ Printable patient reports

INVENTORY & STOCK CONTROL

- ✓ Item & category management
- ✓ Expiry-date monitoring
- ✓ Low-stock alerts & reports
- ✓ Stock adjustment logging
- ✓ FIFO batch tracking
- ✓ Auto-deduction on billing
- ✓ Units of measure
- ✓ AI demand forecasting

BILLING & INVOICING

- ✓ Automated invoice generation
- ✓ Service category management
- ✓ Printable PDF invoices
- ✓ Service & product line items
- ✓ Payment status tracking
- ✓ Emailed receipts



SYSTEM FEATURES

Administration

PART 2 OF 3

REPORTS & ANALYTICS

- ✓ Operations dashboard
- ✓ Sales reports
- ✓ Low-stock reports
- ✓ Analytics & trend charts
- ✓ Patient reports
- ✓ PDF & table export

USERS, ROLES & ACCESS

- ✓ Role-based access control
- ✓ Veterinarian / Staff roles
- ✓ Staff account management
- ✓ Suspend / deactivate / restore
- ✓ Forced password change
- ✓ Super Admin / Clinic Admin
- ✓ Pet Owner portal role
- ✓ Portal user administration
- ✓ Device & session control
- ✓ Google sign-in support

DATA GOVERNANCE & SAFETY

- ✓ Full audit trail logging
- ✓ Soft delete with restore window
- ✓ Bulk data import
- ✓ Species, breeds & weight ranges
- ✓ Archive & recovery bin
- ✓ Database backup & restore
- ✓ Master data management
- ✓ Pet size categories

PLATFORM & CONFIGURATION

- ✓ Clinic profile & branding
- ✓ Maintenance mode
- ✓ System-wide broadcasts
- ✓ Notification templates
- ✓ System preferences
- ✓ Multi-clinic management
- ✓ CMS content pages
- ✓ Client notification history

INTELLIGENCE, REAL-TIME & OFFLINE

- ✓ AI clinical support
- ✓ Live push notifications
- ✓ Queued offline mutations
- ✓ AI inventory forecasting
- ✓ Offline-first operation
- ✓ Automatic cloud sync



SYSTEM FEATURES

Pet Owner Portal

PART 3 OF 3

ACCOUNT & SECURITY

- ✓ Self-service registration
- ✓ Google sign-in
- ✓ Profile management
- ✓ Active device sessions
- ✓ Account deletion request
- ✓ Email verification
- ✓ Forgot & reset password
- ✓ Change / set password
- ✓ Revoke other devices
- ✓ 30-day account recovery

MY PETS

- ✓ Register a new pet
- ✓ Upload pet photo
- ✓ Medical history access
- ✓ Edit pet details
- ✓ View pet profile
- ✓ Vaccination records

APPOINTMENTS

- ✓ Browse available slots
- ✓ View upcoming visits
- ✓ Cancellation requests
- ✓ Book an appointment
- ✓ Appointment history
- ✓ Status & reminder alerts

BILLING & NOTIFICATIONS

- ✓ View clinic invoices
- ✓ Payment status visibility
- ✓ Email notifications
- ✓ Download invoice PDFs
- ✓ In-app notification centre
- ✓ Real-time status updates

ENGAGEMENT & INFORMATION

- ✓ Submit reviews & feedback
- ✓ Privacy policy page
- ✓ Maintenance notices
- ✓ Clinic landing page
- ✓ Terms of service page
- ✓ Account status notices



TECHNOLOGIES USED

AUTOVET TECH STACK



React 19 • TypeScript • Vite • Tailwind CSS

Admin dashboard and pet-owner portal front-ends.



Laravel 12 • PHP 8.2 REST API

Application backend, business rules, and sync bridge.



MySQL • TiDB Cloud

Cloud database with a local MySQL mirror for offline use.



Laravel Sanctum • RBAC • Audit Trail

Token authentication, hashed passwords, per-role permissions.



Laravel Reverb • Pusher • Horizon

Real-time broadcasting and queued background jobs.



Python • Pandas • Scikit-Learn

Regression-based inventory demand forecasting module.



DomPDF • jsPDF • Recharts

Invoice/report PDF generation and analytics charts.



Service Workers • IndexedDB

Offline-first caching and queued mutation replay.



Render • Vercel • Brevo • Sentry

Cloud hosting, static delivery, transactional email, monitoring.

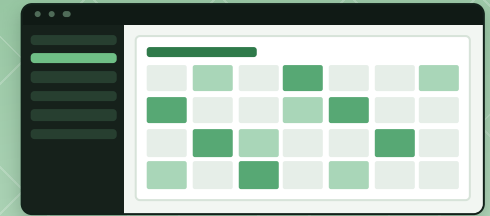


KEY SCREENS

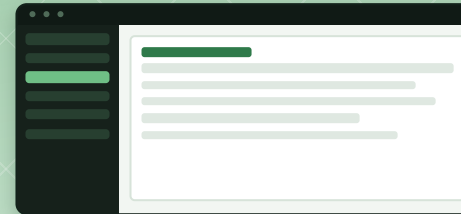
REPRESENTATIVE MODULES



1. ADMIN DASHBOARD



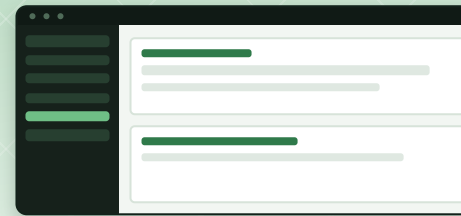
2. APPOINTMENT SCHEDULING



3. INVENTORY & FIFO BATCHES



4. INVOICING & REPORTS



5. PATIENT RECORDS



6. PET OWNER PORTAL